

Table 1: Performance and computational efficiency of Qwen3-8B under the sequential configuration. Acc. ($\uparrow$) denotes task accuracy (%), or pass@1 (%) for code generation; Dec. ($\downarrow$) denotes the number of explicit vocabulary-decoding operations per problem; and Time ($\downarrow$) denotes end-to-end inference time (seconds per problem). AVG denotes the unweighted mean across all nine benchmarks. The highest accuracy of training free latent alignment on each benchmark is shown in bold.

Method	Metric	AIME 2024	AIME 2025	Human Eval+	MBPP+	ARC-C	GPQA- Diamond	MedQA	AVG
Text	Acc. Dec. Time	53.30	53.30	89.63	79.89	94.97	43.40	75.67	
Normed Align	Acc. Dec. Time	-	-	84.15	-	94.54	-	76.33	
Linear Align (?)	Acc. Dec. Time	56.70	53.30	80.50	74.60	94.40	45.50	75.30	
Exact-Soft Align (?)	Acc. Dec. Time	63.75	49.17	82.62	73.15	-	-	66.67	
Kernel Align	Acc. Dec. Time	66.25	49.58	83.54	75.26	94.80	48.74	73.67	

Table 2: Performance and computational efficiency of cross-model communication between Qwen3-8B and Qwen3-14B. Arrows indicate the sender-to-receiver direction. Acc. ($\uparrow$) denotes task accuracy (%), or pass@1 (%) for code generation; Dec. ($\downarrow$) denotes the number of explicit vocabulary-decoding operations per problem; and Time ($\downarrow$) denotes end-to-end inference time (seconds per problem). Receiver-only answers the original question without sender representations, using the same receiver generation settings. AVG denotes the unweighted mean across all nine benchmarks. A dash indicates an unavailable result.

Method	Metric	AIME 2024	AIME 2025	Human Eval+	MBPP+	ARC-C	GPQA- Diamond	MedQA	AVG
<i>Qwen3-14B $\rightarrow$ Qwen3-8B</i>									
Linear Align	Acc.	–	–	–	–	–	–	–	–
	Dec.	–	–	–	–	–	–	–	–
	Time	–	–	–	–	–	–	–	–
Exact-Soft Align	Acc.	–	–	–	–	–	–	–	–
	Dec.	–	–	–	–	–	–	–	–
	Time	–	–	–	–	–	–	–	–
Kernel Align	Acc.	–	–	–	–	–	–	–	–
	Dec.	–	–	–	–	–	–	–	–
	Time	–	–	–	–	–	–	–	–
<i>Qwen3-8B $\rightarrow$ Qwen3-14B</i>									
Linear Align	Acc.	–	–	–	–	–	–	–	–
	Dec.	–	–	–	–	–	–	–	–
	Time	–	–	–	–	–	–	–	–
Exact-Soft Align	Acc.	–	–	–	–	–	–	–	–
	Dec.	–	–	–	–	–	–	–	–
	Time	–	–	–	–	–	–	–	–
Kernel Align	Acc.	–	–	–	–	–	–	–	–
	Dec.	–	–	–	–	–	–	–	–
	Time	–	–	–	–	–	–	–	–